Quantum Entanglement

Leonard Susskind

Supplementary track, 2006 Fall Part 3
Transcript-derived notes curated by LazyingArt LLC

Original lectures by Leonard Susskind. Transcript-derived course notes curated by LazyingArt LLC.

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Chapter 1

Special Relativity: Frames, Simultaneity, and the Lorentz Transformation

These notes follow Leonard Susskind's lecture, with curation for this edition by LazyingArt LLC. The lecture opens by setting the agenda for the quarter: first special relativity done carefully as kinematics, then relativistic dynamics, and only later a more conceptual pass through general relativity. We therefore begin where the lecture itself begins in earnest, with the meaning of a frame of reference, with clocks and meter sticks, and with the insistence that the right way to start a relativity problem is to draw spacetime.

1.1 A Short Framing Note

Relativity, in the sense used here, is the statement that the laws of physics should not depend on the uniform motion of the observer. That idea is older than Einstein. Susskind likes to make the point with a train moving so smoothly that nothing inside it rattles or shakes. In such a train, the laws of juggling are the same as they are in a train standing at the station. If we throw a ball upward, we do not have to compensate for the train's motion in order to catch it again. The physics inside the smoothly moving train is the same physics.

What Einstein changes is not the relativity principle itself, but the structure of space and time needed to preserve it once light is brought into the story. Newton imagined a universal time, almost a God's time, common to all clocks everywhere. Einstein keeps the relativity principle and abandons the universality of simultaneity. The lecture proceeds exactly in that order: first Galileo, then the crisis created by light, then the reconstruction of space and time.

1.2 Reference Frames as Lattices of Meter Sticks and Clocks

Definition 1.1. A reference frame is an imaginary lattice of meter sticks filling space, with a clock at each lattice point. An event is a point in spacetime labeled by a spatial location in that lattice and by the reading of the appropriate clock.

Figure 1.1: Rest-frame time axis and the emerging spacetime lattice.

The lecture immediately turns this operational definition into a picture. Time is drawn vertically, space horizontally, and for the moment we suppress the y and z directions and work only with one spatial coordinate x. Every point on the board then represents a place together with a time.

The horizontal axis is the set of events with $t = 0$. The vertical line through the origin is the worldline of the point $x = 0$. The next vertical line is the worldline of the point $x = 1$, then $x = 2$, and so on. As the clocks tick upward in time, the meter-stick lattice produces a family of constant-position worldlines. When we add horizontal lines of constant time, the whole picture becomes a spacetime lattice.

The board at this point is still sparse, and that is important. Susskind is not yet racing toward formulas. He is teaching us how to think about a frame: as a grid of locations and synchronized clocks, made visible in spacetime before any transformation law is written down.

1.3 Galilean Relativity from the Spacetime Picture

The lecture now brings back the train picture and makes it geometrical. Let the stationary observer use coordinates (x, t), and let the moving observer ride in a train whose origin moves with constant velocity v. In the stationary frame the moving origin has the worldline

$$x = vt. \tag{1.1}$$

The next point of the moving lattice traces a parallel line shifted by a fixed amount, then the next one, and so on. In a spacetime plot the moving family of constant-position lines is tilted.

That tilt is the whole content of Galilean kinematics. An event with stationary coordinates (x, t) has, for the moving observer, the same time but a different spatial coordinate, because the moving

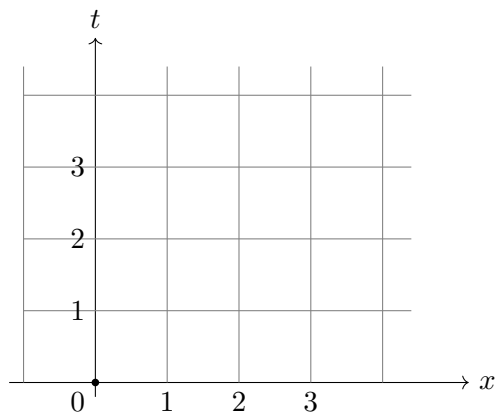

Figure 1.2: A clean reconstruction of the stationary spacetime lattice: constant-position worldlines are vertical and constant-time slices are horizontal.

origin has shifted by the amount vt. Reading directly from the diagram, one arrives at

$$t' = t, \tag{1.2}$$
$$x' = x - vt. \tag{1.3}$$

Susskind is careful about the order here. He does not begin from algebra and then draw a picture afterward. He draws the picture and then reads the equations off the board.

We can invert the relation as well. Since $t = t'$, the inverse form is

$$t = t', \tag{1.4}$$
$$x = x' + vt'. \tag{1.5}$$

The lower inverse equation is cropped in the board frame, and Susskind briefly phrases it as $x = x' + vt$. But because $t = t'$ in Galilean relativity, the intended inverse equation is the canonical one just written.

This is also the point in the lecture where the train analogy and the mathematics meet cleanly. The first pair of equations tells the stationary observer how to convert his description into the moving observer's description; the second pair tells us how to go back. Galilean relativity is, in that sense, a theory of transformations between frames.

1.4 What Galileo Gets Right

Susskind now pauses and asks the right question: what is invariant? He does not want the transformation law to float free of physics.

Take two events at the same time t, with positions x_1 and x_2. In the moving frame,

$$x_1' = x_1 - vt, \tag{1.6}$$
$$x_2' = x_2 - vt. \tag{1.7}$$

Subtracting the two,

$$x_2' - x_1' = (x_2 - vt) - (x_1 - vt) = x_2 - x_1. \tag{1.8}$$

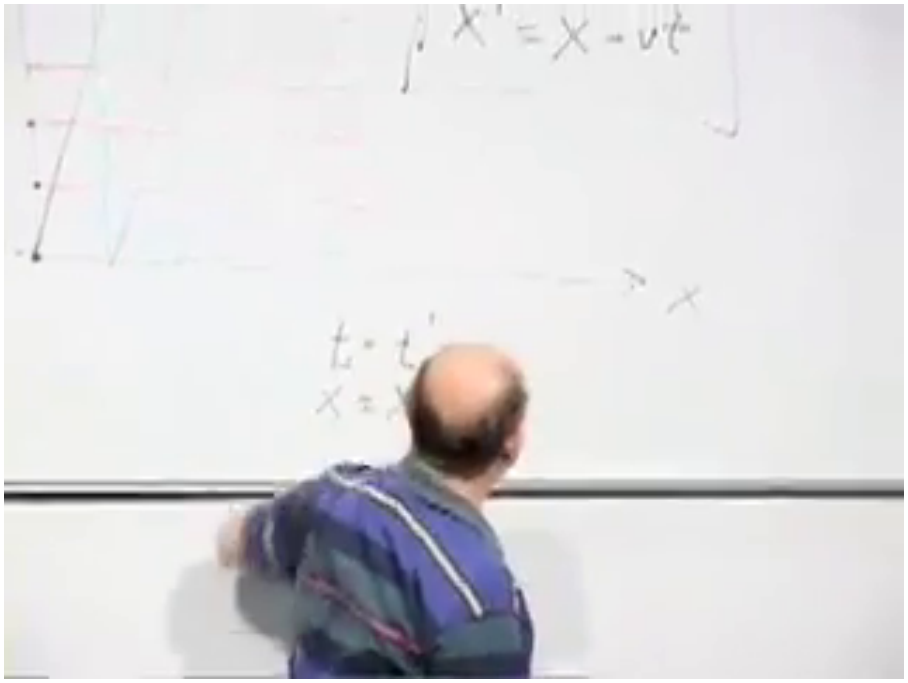

Figure 1.3: Galilean coordinate shift beside the spacetime sketch.

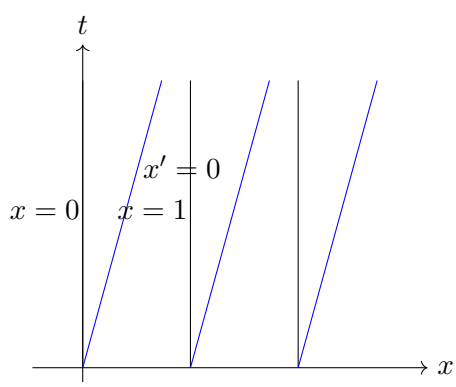

Figure 1.4: Stationary worldlines and the tilted moving family from which the Galilean shift is read.

So the spatial separation of two simultaneous events is invariant under Galilean transformations. The individual coordinates are not invariant, but their difference is.

At this stage the lecture also raises a natural worry. How do we know that time really remains universal when a clock is carried around? Galileo himself did not know in any strict modern sense. It was, as Susskind says, frankly a guess, though a very good one. For ordinary clocks moved around at ordinary speeds, the agreement is fantastically precise. Only at sufficiently high speeds does the guess fail, and that failure is exactly what special relativity is about.

Now let a particle follow a trajectory $x(t)$. The moving observer sees

$$x'(t) = x(t) - vt. \tag{1.9}$$

So position is not invariant. Different observers attach different spatial coordinates to the same particle. Differentiating once,

$$\frac{dx'}{dt} = \frac{dx}{dt} - v, \tag{1.10}$$

and therefore velocity is not invariant either. This is obvious from ordinary experience: if I move along beside your arrow, I judge its speed differently from you.

Differentiate once more:

$$\frac{d^2x'}{dt^2} = \frac{d^2x}{dt^2}, \tag{1.11}$$

because the relative velocity v between inertial frames is constant. Acceleration is invariant.

This is the Newtonian payoff. If the force between two objects depends only on the distance between them, and if that distance is invariant, then the force is also invariant. If acceleration is invariant as well, then

$$F = ma \tag{1.12}$$

has precisely the right structure to serve as a law of motion that all inertial observers can share, provided mass itself is invariant. In this sense Newton's laws are invariant under Galilean coordinate transformations.

The lecture's point is not merely historical courtesy. Galileo and Newton were not making a naive mistake. For mechanics at low velocities, the whole scheme is extraordinarily successful.

1.5 Maxwell's Challenge and Einstein's Return to First Principles

The next move is the crisis. Galilean relativity works beautifully for mechanics, but electromagnetism changes the stakes. Maxwell's equations predict a definite speed for light, and if Maxwell's equations are taken seriously as laws of nature, then the speed of light ought to be the same for every inertial observer.

Galilean kinematics says otherwise. If a light ray in the stationary frame satisfies

$$x = ct, \tag{1.13}$$

then the moving observer would assign

$$x' = x - vt = (c - v)t. \tag{1.14}$$

So under a Galilean transformation the moving observer sees light propagate at speed $c - v$, not at c. That is exactly what one would expect for sound waves or water waves. Indeed Susskind stresses that this was not a crazy expectation. It was the natural expectation.

The historical temptation, before Einstein, was to conclude that there must be a preferred frame, a kind of God's frame, in which Maxwell's equations are exactly right and in which light really travels at speed c. Einstein's move is more radical and more economical. He keeps the relativity principle and forces space and time to adapt. In the lecture this comes in the form of two postulates:

1. The laws of physics are identical in every inertial frame.
2. The speed of light is the same in every inertial frame.

But these postulates do not yet tell us how to assign coordinates. So Einstein goes back to first principles and asks what it means for clocks at different places to be synchronized. The answer is operational. Put an observer at the midpoint between two endpoint clocks. Emit flashes from the endpoints when the endpoint clocks read the same time. If the midpoint observer receives the flashes simultaneously, the clocks are synchronized.

This is the point where the lecture stops being Newtonian in spirit. If a second observer moves past that synchronized apparatus, the two flashes no longer arrive at the moving midpoint at the same instant. The moving observer must therefore conclude that the original emissions were not simultaneous. Simultaneity is not absolute.

1.5.1 Question & Answer

Question. Why not synchronize clocks with sound instead of light?

Answer. One can synchronize clocks with sound in a frame in which the air is at rest. But then one immediately discovers that moving observers measure different sound speeds. Sound propagates in a medium, and motion relative to that medium matters. In an open-air train moving through still air, sound would be a bad messenger of relativity. Light is chosen for the opposite reason: Einstein's postulate is that light has the same speed for every inertial observer, so light is exactly the signal with which the notion of simultaneity must be rebuilt.

The lecture now asks its central quantitative question: if the stationary observer calls one pair of distant events simultaneous, which pair does the moving observer call simultaneous instead?

1.6 The Geometry of Relativity of Simultaneity

Susskind now slows down and makes the question geometric. He sets $c = 1$, so that space and time are measured in compatible units and light rays run at 45°. A right-moving light ray has the form

$$x = t + d, \tag{1.15}$$

and a left-moving light ray has the form

$$x = -t + d'. \tag{1.16}$$

In particular, the light ray from the origin satisfies $x = t$.

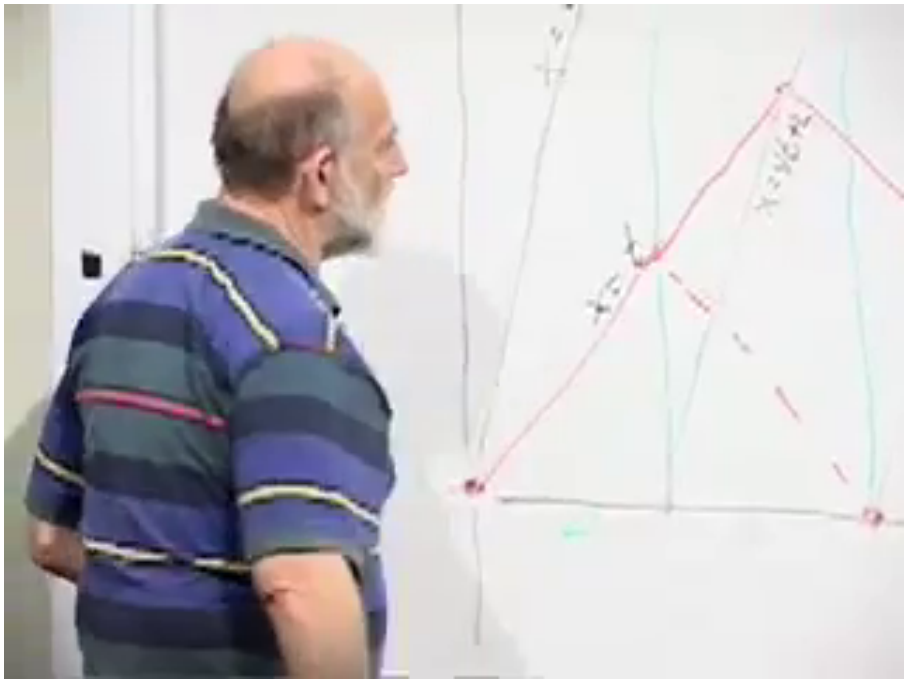

Figure 1.5: Light-ray line $x = t$ in the simultaneity construction.

He then lays out the rest-frame synchronization experiment with endpoints separated by equal distances L. The midpoint observer sits between them. Once the moving frame is added, the question becomes concrete: which event on the right-moving family is simultaneous, in the moving sense, with the event at the origin?

On the board only the label $x = t$ is clearly legible. The shifted moving worldlines are reconstructed from the transcript and from the geometry as

$$x = vt, \qquad x = vt + L, \qquad x = vt + 2L. \tag{1.17}$$

One transcript line later drops the factor of t, but the construction itself makes the intended form unambiguous.

The lecture's rhythm at this point is exact: it first argues qualitatively that the two frames will disagree, and then asks, "Can we do it quantitatively?" The answer is yes.

Worked derivation. Let A be the point where the forward light ray $x = t$ meets the shifted worldline $x = vt + L$. Then A is determined by

$$x = t, \tag{1.18}$$

$$x = vt + L. \tag{1.19}$$

Substituting $x = t$ into the second equation gives

$$t = vt + L, \tag{1.20}$$

so

$$(1 - v)t = L, \tag{1.21}$$

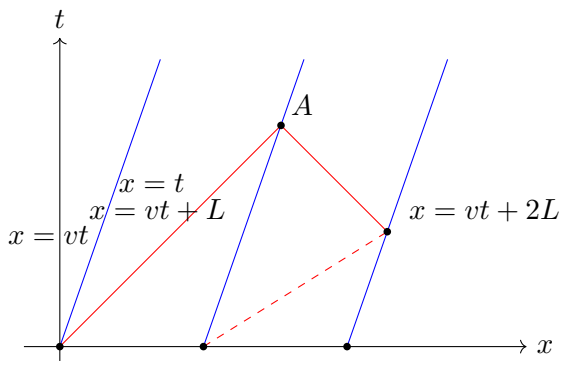

Figure 1.6: Reconstruction of the simultaneity argument. The screenshot preserves the board evidence; the redraw makes the equations explicit.

and therefore

$$t_A = \frac{L}{1-v}, \qquad x_A = \frac{L}{1-v}. \tag{1.22}$$

Now follow a backward-going light ray from A. A left-moving light ray has the form

$$x = -t + b, \tag{1.23}$$

or equivalently

$$x + t = b. \tag{1.24}$$

To determine b, use the fact that the ray passes through A. At A, we have $x = t = L/(1-v)$, so

$$b = x_A + t_A = \frac{2L}{1-v}. \tag{1.25}$$

Thus the backward light ray is

$$x + t = \frac{2L}{1-v}. \tag{1.26}$$

The event we want is where this backward light ray meets the next shifted worldline $x = vt + 2L$. So we solve

$$x + t = \frac{2L}{1-v}, \tag{1.27}$$

$$x = vt + 2L. \tag{1.28}$$

Substituting the second equation into the first yields

$$vt + 2L + t = \frac{2L}{1-v}. \tag{1.29}$$

Rearranging,

$$(1+v)t = 2L\left(\frac{1}{1-v} - 1\right) = 2L\frac{v}{1-v}, \tag{1.30}$$

and therefore

$$t = \frac{2Lv}{1-v^2}. \tag{1.31}$$

Putting this back into $x = vt + 2L$, we obtain

$$x = \frac{2L}{1 - v^2}. \tag{1.32}$$

What matters is not these coordinates by themselves, but their ratio:

$$\frac{t}{x} = v. \tag{1.33}$$

So the moving observer's simultaneity line through the origin is

$$t = vx. \tag{1.34}$$

This is the lecture's geometric payoff. The moving observer's time axis is

$$x = vt, \tag{1.35}$$

because that is the worldline of the moving origin. The moving observer's line of simultaneity is

$$t = vx, \tag{1.36}$$

because those are the points assigned the same moving time. The two are tilted symmetrically with respect to the light rays.

Restoring dimensions requires only one correction:

$$t = \frac{v}{c^2}x. \tag{1.37}$$

This formula makes clear why simultaneity appears absolute in ordinary life. The denominator c^2 is enormous. Only when velocities become appreciable compared with the speed of light does the tilt become visible.

1.7 From Tilted Axes to the Lorentz Transformation

The lecture now performs an instructive cleanup. It erases the clutter of the construction and keeps only what the construction has taught us: the unprimed axes, the primed axes, and the light rays. In units with $c = 1$, the picture has what Susskind calls a lovely symmetry.

The lecture later shifts on the board to upper-case X, T, but the substance is unchanged. We keep lower-case x, t throughout.

The geometry already tells us the zero-sets of the primed coordinates. Since the moving origin is the line $x = vt$, the quantity x' must vanish there. That is the surviving content of Galileo's old formula $x' = x - vt$: even if the exact coefficient changes, the zero-set must remain the same. This motivates

$$x' = \frac{x - vt}{F(v)}. \tag{1.38}$$

Likewise, because the moving simultaneity axis is the line $t = vx$, the quantity t' must vanish there. So we write

$$t' = \frac{t - vx}{G(v)}. \tag{1.39}$$

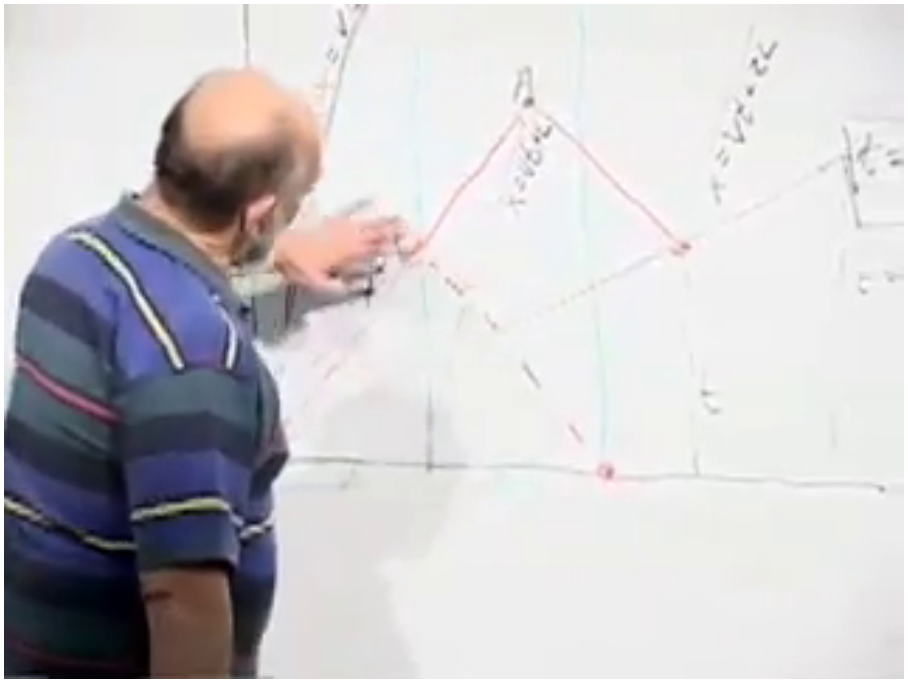

Figure 1.7: Symmetric $c = 1$ spacetime diagram.

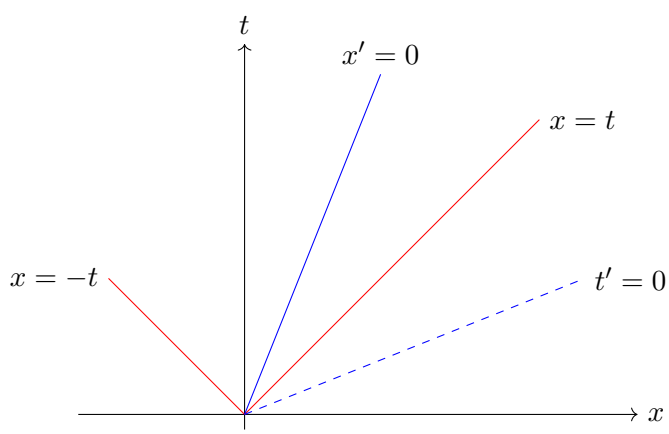

Figure 1.8: Cleaned $c = 1$ diagram showing the primed time axis, the primed simultaneity axis, and the two light rays.

Here $F(v)$ and $G(v)$ are unknown denominator functions. This matters: when the lecture later concludes that $F(v) = \sqrt{1-v^2}$, it is the denominator that equals that square root.

Now use the invariance of light speed. A light ray satisfying $x = t$ in the unprimed frame must satisfy $x' = t'$ in the primed frame. Substituting $x = t$ gives

$$x' = \frac{(1-v)t}{F(v)}, \tag{1.40}$$

$$t' = \frac{(1-v)t}{G(v)}. \tag{1.41}$$

These must agree for every such light ray, so

$$F(v) = G(v). \tag{1.42}$$

Hence

$$x' = \frac{x - vt}{F(v)}, \qquad t' = \frac{t - vx}{F(v)}. \tag{1.43}$$

We still do not know $F(v)$. To determine it, multiply through:

$$F(v)x' = x - vt, \tag{1.44}$$

$$F(v)t' = t - vx. \tag{1.45}$$

To solve for x, multiply the second equation by v and add it to the first. The vt terms cancel:

$$(1-v^2)x = F(v)(x' + vt'), \tag{1.46}$$

so

$$x = \frac{F(v)(x' + vt')}{1-v^2}. \tag{1.47}$$

By the same algebra,

$$t = \frac{F(v)(t' + vx')}{1-v^2}. \tag{1.48}$$

Now comes what Susskind calls the other half of the key. The forward and inverse transformations should have the same form, except that the sign of the relative velocity reverses. This reciprocity condition demands

$$\frac{F(v)}{1-v^2} = \frac{1}{F(v)}. \tag{1.49}$$

Multiplying through by $F(v)$, we obtain

$$F(v)^2 = 1 - v^2, \tag{1.50}$$

and therefore

$$F(v) = \sqrt{1-v^2}. \tag{1.51}$$

The Lorentz transformation in units with $c = 1$ is now fixed:

$$x' = \frac{x - vt}{\sqrt{1-v^2}}, \tag{1.52}$$

$$t' = \frac{t - vx}{\sqrt{1-v^2}}. \tag{1.53}$$

By reciprocity, the inverse relations are

$$x = \frac{x' + vt'}{\sqrt{1 - v^2}}, \tag{1.54}$$

$$t = \frac{t' + vx'}{\sqrt{1 - v^2}}. \tag{1.55}$$

To restore dimensions, we replace v^2 by v^2/c^2, and in the time equation we divide the mixed term by c^2:

$$x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}}, \tag{1.56}$$

$$t' = \frac{t - (v/c^2)x}{\sqrt{1 - v^2/c^2}}. \tag{1.57}$$

If one later introduces

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}, \tag{1.58}$$

these become the familiar textbook formulas. But the lecture reaches them first in denominator form, and it is worth keeping that sequence.

A direct check. The lecture closes by checking the defining property of the transformation. If

$$x = ct, \tag{1.59}$$

then

$$x' = \frac{(c - v)t}{\sqrt{1 - v^2/c^2}}, \tag{1.60}$$

$$t' = \frac{(1 - v/c)t}{\sqrt{1 - v^2/c^2}}. \tag{1.61}$$

Hence

$$\frac{x'}{t'} = \frac{c - v}{1 - v/c} = c. \tag{1.62}$$

So a right-moving light ray still travels with speed c in the primed frame. The same computation shows that a left-moving light ray satisfying $x = -ct$ becomes $x' = -ct'$.

The lecture also insists on the low-velocity limit. When $v \ll c$, the factors v^2/c^2 and $(v/c^2)x$ are negligible in ordinary circumstances, and the Lorentz transformation reduces to

$$t' \approx t, \qquad x' \approx x - vt. \tag{1.63}$$

Galileo is recovered as an approximation. But the new thing, the genuinely bizarre thing in Susskind's phrasing, is already visible: the surfaces $t =$ constant and $t' =$ constant are different surfaces. Simultaneity is frame-dependent.

1.8 Summary

This lecture unfolds in a deliberate order. We begin with a reference frame understood operationally as a lattice of meter sticks and synchronized clocks. We draw spacetime before we write equations. From that geometry we read the Galilean transformation and then test it: simultaneous distance is invariant, position and velocity are not, acceleration is, and so Newtonian mechanics acquires the right invariant form.

Only after Galileo has been allowed to look strong does the real problem appear. Maxwell and light do not fit Galilean kinematics. Einstein's response is not to abandon relativity, but to return to first principles and rebuild synchronization with light signals. From that reconstruction comes the relativity of simultaneity.

The concrete light-ray construction then gives the moving time axis $x = vt$ and the moving simultaneity axis $t = vx$ in units with $c = 1$. From there, plus the invariance of light speed and the reciprocity of the two frames, the Lorentz transformation follows. The lecture stops before length contraction and time dilation are derived. Its real achievement is more basic: it has shown us that time is not simply time, and that what counts as synchronous in one frame is not what counts as synchronous in another.

Chapter 2

Maxwell Equations, Covariance, and Electromagnetic Waves

We begin exactly where the previous lecture stopped. We already know how electric and magnetic fields act on charged particles; the Lorentz force law tells us that part of the story. The missing half is the reverse question: what determines the fields themselves, and how do charges and currents generate them? In this lecture Leonard Susskind answers that question in a deliberately relativistic way. Maxwell's equations are not introduced merely as a list to memorize, but as field equations to be motivated physically, written covariantly, checked component by component, and then used to derive the existence of light waves. Throughout we work in units with $c = 1$. These notes follow Leonard Susskind closely and are curated by LazyingArt LLC.

2.1 From the Lorentz Force to the Missing Field Equations

The lecture opens by looking backward before it moves forward. Electric and magnetic fields push charged particles around, and the force is proportional to the charge because an uncharged particle does not feel the electromagnetic field at all. In ordinary three-vector language the law is

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \tag{2.1}$$

This is the familiar half of electromagnetism: given the fields, we know how matter responds.

The board then pivots toward covariant notation. As written there, the second line is only schematic; it suppresses the free force index and the contraction with the four-velocity. The clean covariant statement is

$$f^\mu = q\, F^\mu{}_\nu\, u^\nu. \tag{2.2}$$

Here f^μ is the four-force, u^ν is the four-velocity, and $F^{\mu\nu}$ is the antisymmetric electromagnetic field tensor that we will reconstruct shortly. The point of the reminder is not just formal prettiness. Written this way, the Lorentz force law is manifestly the same in every inertial frame.

That is already the standard against which Maxwell's equations will be judged. The missing question is no longer how a charged particle moves in a prescribed field. The missing question is what determines the field itself: what decides whether there is an electric field here, a magnetic field

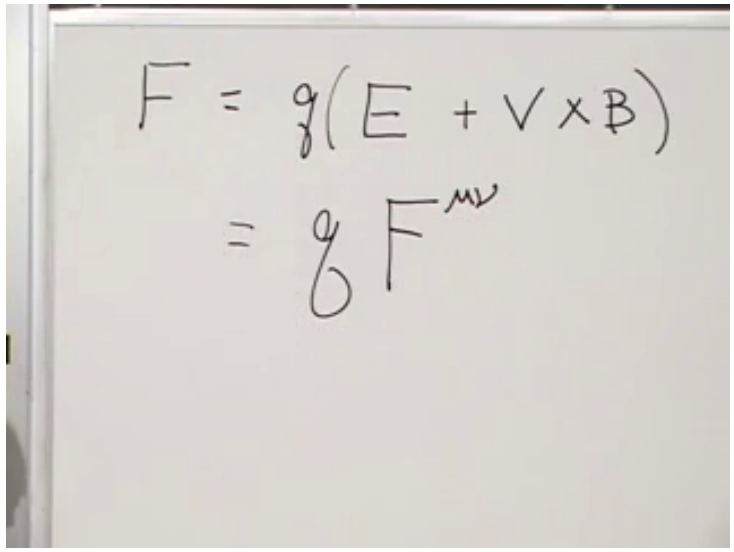

Figure 2.1: Lorentz force rewritten schematically in field-tensor notation.

there, or an electromagnetic wave moving through empty space? Those are the equations we now need, and those equations are Maxwell's equations.

As always in this course, the broader issue is relativity. If Maxwell's equations are the theory of light, and if the laws of physics are the same in all inertial frames, then light itself must come out with the same speed in all inertial frames. That is the puzzle Einstein struggled with, and the lecture keeps it in view from the start.

2.2 Why Charges Must Affect Fields

Before writing any field equations, Susskind slows down and asks why charged matter must affect the field at all. A point charge creates an electric field spreading outward from it. A current in a wire creates a magnetic field circulating around the wire. Those are familiar empirical facts, but the lecture wants the deeper logic that makes Maxwell's equations necessary.

The key point is reciprocity sharpened by conservation laws. An electromagnetic wave can hit a charged particle, change its motion, and give it energy. If that can happen, then the influence cannot run only one way. Charged matter must also be able to alter the electromagnetic field. Otherwise energy could appear in the particle without being taken from anywhere.

2.2.1 Question & Answer

Question. Why must charged matter act back on the electromagnetic field?

Answer. Because otherwise energy conservation would fail. If a field can do work on a charge, then a charge must be able to draw energy from the field and modify it. One may call this action and reaction, but the lecture's sharper point is conservation: the field cannot give energy to matter unless matter can affect the field in return.

The lecture then turns that abstract point into a causal picture. Suppose we move a point charge suddenly from one place to another. The field cannot rearrange itself instantaneously throughout all of space. A distant observer is not allowed to see the effect before a light signal could have arrived. Thus the field must rearrange by sending out a disturbance. Near the charge, the rearrangement happens first; farther away, the old field remains until the signal reaches it. If we shake the charge back and forth, the outward-moving disturbance becomes an electromagnetic wave. An antenna is precisely such a device.

The same story can be told for currents. Reverse a current in a wire, and the magnetic field around the wire must reverse as well. But that reversal cannot be known arbitrarily far away in arbitrarily short time. The change in the magnetic field must propagate outward, again as an electromagnetic disturbance. This is the basic physics Maxwell's equations must describe.

2.3 Wave Review and Vector-Calculus Review

Only now does the lecture pause for review. This is not filler. Before Maxwell's equations can be made to produce waves, we first recall what wave equations look like.

Start with a scalar field ϕ, a single number attached to each point of spacetime. The relativistic wave equation is

$$\partial_\mu \partial^\mu \phi = 0. \tag{2.3}$$

With the sign convention used throughout the lecture, this means

$$\partial_t^2 \phi - \partial_x^2 \phi - \partial_y^2 \phi - \partial_z^2 \phi = 0. \tag{2.4}$$

If the wave moves along the z-axis, then ϕ depends on z and t but not on x or y, and the equation reduces to

$$\partial_t^2 \phi - \partial_z^2 \phi = 0. \tag{2.5}$$

The lecture emphasizes the two basic traveling-wave solutions,

$$\phi = f(z - t), \qquad \phi = g(z + t), \tag{2.6}$$

representing right-moving and left-moving profiles. A particularly useful special case is the sinusoidal wave

$$\phi(z, t) = \cos\big(k(z - t)\big), \tag{2.7}$$

where k is the wave number. Large k means short wavelength; small k means long wavelength. These are exactly the shapes that will later reappear in the electric and magnetic fields themselves.

The lecture then reminds us of the vector operations Maxwell's equations use. For ordinary three-vectors,

$$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z, \tag{2.8}$$

$$(\mathbf{a} \times \mathbf{b})_x = a_y b_z - a_z b_y, \tag{2.9}$$

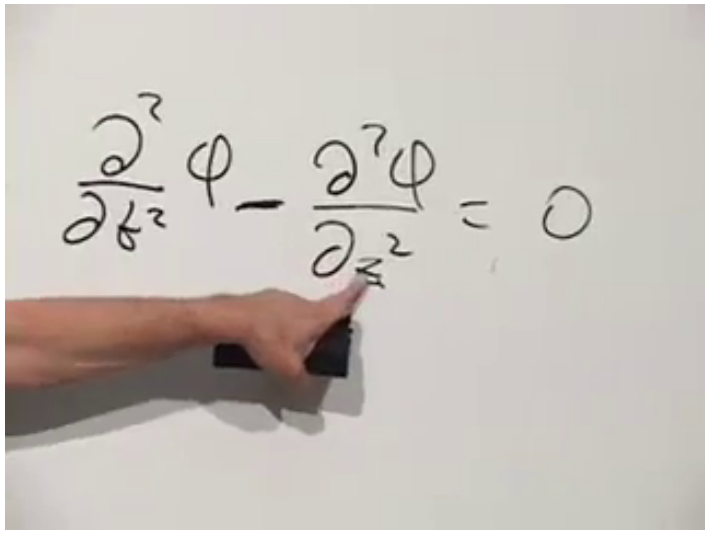

Figure 2.2: One-dimensional reduction of the wave equation.

with the remaining components obtained by cyclic permutation. For a vector field $\mathbf{A}$,

$$\nabla \cdot \mathbf{A} = \partial_x A_x + \partial_y A_y + \partial_z A_z, \tag{2.10}$$

$$(\nabla \times \mathbf{A})_x = \partial_y A_z - \partial_z A_y, \tag{2.11}$$

again with cyclic permutation for the other components.

Susskind keeps the geometry in sight. Divergence measures the extent to which a field seems to emanate from somewhere. Curl measures the extent to which it circulates around something. A field of arrows flowing outward from a point suggests divergence; a field winding around a wire suggests curl. These are the operational meanings needed when Maxwell's equations are written down.

2.4 Vacuum Maxwell Equations and the Relativity Puzzle

With that preparation in place, the lecture writes the source-free Maxwell equations, in the order actually used on the board:

$$\nabla \times \mathbf{E} = -\,\partial_t \mathbf{B}, \tag{2.12}$$

$$\nabla \cdot \mathbf{B} = 0, \tag{2.13}$$

$$\nabla \times \mathbf{B} = \partial_t \mathbf{E}, \tag{2.14}$$

$$\nabla \cdot \mathbf{E} = 0. \tag{2.15}$$

These are the vacuum Maxwell equations. They describe electric and magnetic fields in the absence of charge density and current density.

Two features are immediately visible. First, the curl equations are almost symmetric under interchange of $\mathbf{E}$ and $\mathbf{B}$, but not quite: there is an essential sign difference. Second, the divergence equations are asymmetric. The electric field will later acquire a source term ρ, but the magnetic field does not acquire an analogous source term in this lecture, so $\nabla \cdot \mathbf{B} = 0$.

Susskind briefly points out where sources will eventually go. One will later write $\nabla \cdot \mathbf{E} = \rho$ and $\nabla \times \mathbf{B} = \partial_t \mathbf{E} + \mathbf{J}$, but for the moment those terms are set aside. This postponement is deliberate. The vacuum equations reveal the covariant structure in its cleanest form.

2.4.1 Question & Answer

Question. How can Maxwell's equations be the same in every frame if they imply the same light speed in every frame?

Answer. That is precisely the relativistic puzzle. If Maxwell's equations are the theory of light, and if those equations are the same in every inertial frame, then every inertial frame must predict the same light speed from them. The resolution is not to patch the wave speed by hand from frame to frame. The resolution is to express the laws in terms of objects with definite Lorentz transformation rules, so that the equations themselves are manifestly frame-independent.

That is why the lecture now turns to the electromagnetic field tensor. The payoff is not elegance by itself. The payoff is that once the equations are written as tensor equations, their invariance is visible, and the constancy of the speed of light follows from the same structure.

2.5 The Electromagnetic Field Tensor

The next move is the central formal pivot of the lecture. The electric and magnetic fields are assembled into a single antisymmetric 4×4 tensor $F^{\mu\nu}$. The time coordinate is placed first, so the row and column order is t, x, y, z. The board image fixes the electric entries and the antisymmetric layout.

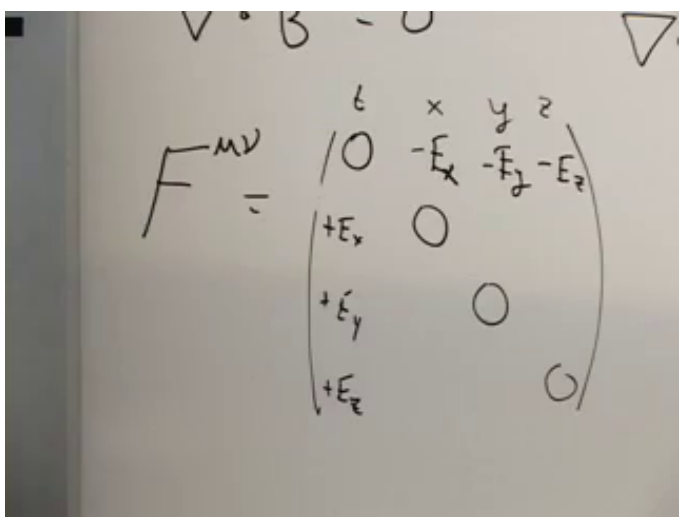

Figure 2.3: Board layout of the electromagnetic field tensor.

With the sign convention fixed by the visible board entries, we write

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & -B_z & B_y \\ E_y & B_z & 0 & -B_x \\ E_z & -B_y & B_x & 0 \end{pmatrix}. \tag{2.16}$$

The diagonal entries vanish because the tensor is antisymmetric, and the entries below the diagonal are the negatives of those above it. The time-space slots contain the electric field; the purely spatial slots contain the magnetic field.

The lecture then introduces a second antisymmetric tensor, the dual tensor $\tilde{F}^{\mu\nu}$. We do not need to overdevelop it. What matters is the replacement rule, stated cautiously because Susskind corrects himself while talking through the signs:

$$\mathbf{E} \mapsto -\mathbf{B}, \qquad \mathbf{B} \mapsto \mathbf{E}. \tag{2.17}$$

With that convention, $\tilde{F}^{\mu\nu}$ packages the other half of Maxwell's equations.

At this stage the lecture's rhythm is very clear: write the tensor, inspect its slots, and then ask what covariant first-derivative equations it could possibly satisfy. Since Maxwell's equations contain only first derivatives and should leave one free index, the natural guess is

$$\partial_\mu F^{\mu\nu} = 0, \tag{2.18}$$

$$\partial_\mu \tilde{F}^{\mu\nu} = 0. \tag{2.19}$$

There are four equations in the first line and four in the second, one for each choice of ν.

Now comes the trust-building step: we check components. For the time component,

$$\partial_\mu F^{\mu t} = \partial_t F^{tt} + \partial_x F^{xt} + \partial_y F^{yt} + \partial_z F^{zt} \tag{2.20}$$

$$= \partial_x E_x + \partial_y E_y + \partial_z E_z \tag{2.21}$$

$$= \nabla \cdot \mathbf{E}. \tag{2.22}$$

Thus $\partial_\mu F^{\mu t} = 0$ is exactly $\nabla \cdot \mathbf{E} = 0$.

To stay close to the lecture, it is also useful to check a spatial component. Taking $\nu = z$,

$$\partial_\mu F^{\mu z} = \partial_t F^{tz} + \partial_x F^{xz} + \partial_y F^{yz} + \partial_z F^{zz} \tag{2.23}$$

$$= -\partial_t E_z + \partial_x B_y - \partial_y B_x \tag{2.24}$$

$$= -\partial_t E_z + (\nabla \times \mathbf{B})_z. \tag{2.25}$$

So $\partial_\mu F^{\mu z} = 0$ is precisely the z-component of

$$\nabla \times \mathbf{B} = \partial_t \mathbf{E}. \tag{2.26}$$

The other spatial components work the same way. Likewise, $\partial_\mu \tilde{F}^{\mu\nu} = 0$ packages $\nabla \cdot \mathbf{B} = 0$ together with $\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$.

This is the formal heart of the lecture. Once Maxwell's equations are written as tensor equations, we can see immediately that they are the same in every inertial frame.

2.6 From Maxwell to Electromagnetic Waves

The lecture then pivots from invariance to content: now that Maxwell's equations are written covariantly, what do they actually say? The answer is that they allow wave-like solutions.

We begin with

$$\nabla \times \mathbf{B} = \partial_t \mathbf{E} \tag{2.27}$$

and differentiate once more with respect to time:

$$\partial_t^2 \mathbf{E} = \nabla \times \partial_t \mathbf{B}. \tag{2.28}$$

Now we use the other curl equation,

$$\partial_t \mathbf{B} = -\nabla \times \mathbf{E}, \tag{2.29}$$

to eliminate $\mathbf{B}$. This gives

$$\partial_t^2 \mathbf{E} = -\nabla \times (\nabla \times \mathbf{E}). \tag{2.30}$$

So far the equations still involve the whole vector $\mathbf{E}$. The next step, exactly as in the lecture, is to isolate one component, say E_x.

The board algebra is long, but its strategy is simple. The x-component of the double curl can be reorganized into a clean form:

$$\partial_t^2 E_x = -\big(\nabla \times (\nabla \times \mathbf{E})\big)_x \tag{2.31}$$

$$= -\partial_x(\nabla \cdot \mathbf{E}) + \partial_x^2 E_x + \partial_y^2 E_x + \partial_z^2 E_x. \tag{2.32}$$

In vacuum, $\nabla \cdot \mathbf{E} = 0$, so the first term drops out and we obtain

$$\partial_t^2 E_x - \partial_x^2 E_x - \partial_y^2 E_x - \partial_z^2 E_x = 0. \tag{2.33}$$

Thus the x-component of the electric field satisfies the same wave equation that we earlier wrote for the scalar field ϕ. This is the exact mathematical payoff Susskind wants. The electric field itself can propagate as a wave.

A simple plane-wave example is now immediate:

$$E_x(z,t) = \cos(z-t), \qquad E_y = 0. \tag{2.34}$$

This is a right-moving wave along the z-axis in the lecture's units. It is also polarized: the electric field has been chosen to point along the x-direction. With the sign convention fixed above, a consistent accompanying magnetic field is

$$B_y(z,t) = \cos(z-t), \qquad B_x = B_z = 0. \tag{2.35}$$

This explicit B_y is not copied from the board; it is the clean consequence of the chosen sign convention together with $\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$.

2.6.1 Question & Answer

Question. Why must the electromagnetic wave be transverse?

Answer. Because the divergence and curl equations leave no room for a propagating longitudinal component. For a wave depending only on z and t,

$$\nabla \cdot \mathbf{E} = \partial_x E_x + \partial_y E_y + \partial_z E_z = \partial_z E_z. \tag{2.36}$$

In vacuum this must vanish, so

$$\partial_z E_z = 0. \tag{2.37}$$

Therefore E_z can at most be a constant. A constant field is not part of the wave profile, so the wave part of the z-component must be absent.

Now look at the curl. If $E_x(z,t)$ is nonzero while $E_y = 0$ and the wave part of E_z vanishes, then the only nonzero component of $\nabla \times \mathbf{E}$ lies along y. But $\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$, so the magnetic field itself must point along y. Hence

$$\mathbf{E} \perp \hat{z}, \qquad \mathbf{B} \perp \hat{z}, \qquad \mathbf{E} \perp \mathbf{B}. \tag{2.38}$$

That is the transversality of electromagnetic radiation. The electric field, the magnetic field, and the direction of propagation are mutually perpendicular.

Remark 2.1. The next frame is kept because it preserves the board geometry of the wave, even though its subtitle metadata is mismatched. Its evidentiary value here is visual: it belongs to the wave-and-polarization discussion, not to the anti-symmetric-matrix discussion suggested by the subtitle timing.

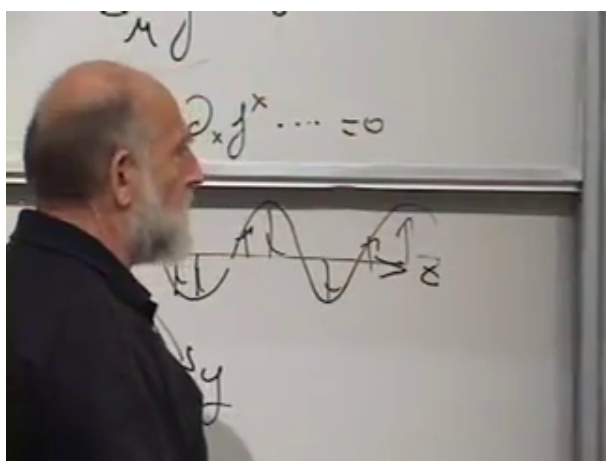

Figure 2.4: Board sketch of a transverse wave propagating along z.

This is the lecture's physical climax. Maxwell's equations imply electromagnetic waves, those waves move with speed 1 in the chosen units, and their electric and magnetic fields are transverse and polarized. Because the equations themselves are the same in every frame, these structural properties belong to every inertial observer.

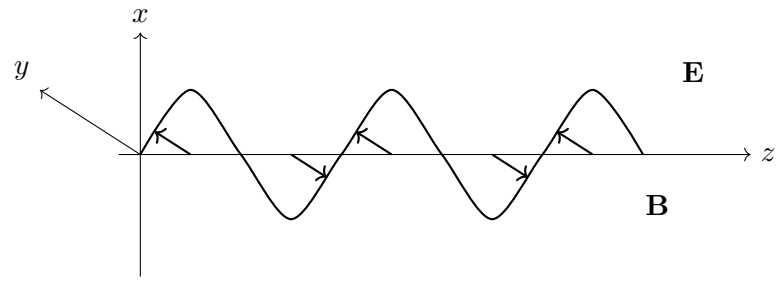

Figure 2.5: A clean reconstruction of the transverse electromagnetic-wave geometry: propagation along z, electric field along x, magnetic field along y.

2.7 Restoring Sources: Charge, Current, Continuity, and Covariance

Only after the vacuum story is clear does the lecture restore what had been postponed: charge density and current density. The Maxwell equations become

$$\nabla \cdot \mathbf{E} = \rho, \tag{2.39}$$

$$\nabla \cdot \mathbf{B} = 0, \tag{2.40}$$

$$\nabla \times \mathbf{E} = -\,\partial_t \mathbf{B}, \tag{2.41}$$

$$\nabla \times \mathbf{B} = \partial_t \mathbf{E} + \mathbf{J}. \tag{2.42}$$

Here ρ is the charge density and $\mathbf{J}$ is the current density.

The charge density is defined by

$$\rho = \frac{dQ}{dV}, \tag{2.43}$$

so that the charge in a region V is

$$Q_V = \int_V \rho \, dV. \tag{2.44}$$

The lecture emphasizes the continuum approximation: in reality space may contain many individual charges, but if we coarse-grain over sufficiently small regions we may treat charge as a continuous density field.

The density ρ can depend on time as well as position, because charge can move. But it cannot change arbitrarily. The total amount of charge in the universe is conserved, and the lecture immediately strengthens that statement. Charge is not allowed simply to disappear here and reappear there with nothing in between. If charge leaves one region and appears in another, there must be a flow of charge connecting the two.

2.7.1 Question & Answer

Question. Why is charge conservation stronger than the statement that total charge is constant?

Answer. Because global constancy by itself would still allow absurd behavior: charge could vanish on Earth and reappear on Mars without anything happening in between. The lecture forbids that. Charge may move from one place to another only by flowing through the intervening space. That stronger, local form of conservation is what leads to current density and to the continuity equation.

To describe such flow, take an infinitesimal oriented surface element $d\boldsymbol{\sigma}$. The charge per unit time passing through it is

$$\frac{dQ}{dt} = \mathbf{J} \cdot d\boldsymbol{\sigma}. \tag{2.45}$$

This is why current is a vector. Its magnitude is charge per unit area per unit time, and its direction is the direction in which charge flows.

Now fix a spatial region V with outward-pointing boundary element $d\boldsymbol{\sigma}$. If net current flows outward through the boundary, the charge inside must decrease. Thus

$$\frac{dQ_V}{dt} = -\oint_{\partial V} \mathbf{J} \cdot d\boldsymbol{\sigma}. \tag{2.46}$$

Since $Q_V = \int_V \rho \, dV$, we also have

$$\frac{dQ_V}{dt} = \int_V \partial_t \rho \, dV. \tag{2.47}$$

Combining the two,

$$\int_V \partial_t \rho \, dV = -\oint_{\partial V} \mathbf{J} \cdot d\boldsymbol{\sigma}. \tag{2.48}$$

At this point the lecture invokes Gauss's theorem, also called the divergence theorem:

$$\oint_{\partial V} \mathbf{A} \cdot d\boldsymbol{\sigma} = \int_V (\nabla \cdot \mathbf{A}) \, dV. \tag{2.49}$$

Applying it to $\mathbf{J}$ gives

$$\int_V \partial_t \rho \, dV = -\int_V (\nabla \cdot \mathbf{J}) \, dV. \tag{2.50}$$

If this holds for every region V, then the integrands themselves must agree, and we obtain

$$\partial_t \rho + \nabla \cdot \mathbf{J} = 0. \tag{2.51}$$

This is the continuity equation, the local form of charge conservation.

Now the relativistic packaging becomes natural. The four quantities ρ, J_x, J_y, J_z form the four-current

$$J^\mu = (\rho, J_x, J_y, J_z), \tag{2.52}$$

and the continuity equation becomes

$$\partial_\mu J^\mu = 0. \tag{2.53}$$

The sourced Maxwell equations themselves then collapse into a single covariant equation:

$$\partial_\mu F^{\mu\nu} = J^\nu. \tag{2.54}$$

For $\nu = t$, the right-hand side is $J^t = \rho$, so the equation becomes

$$\nabla \cdot \mathbf{E} = \rho. \tag{2.55}$$

For spatial ν, it becomes

$$\nabla \times \mathbf{B} = \partial_t \mathbf{E} + \mathbf{J}. \tag{2.56}$$

The other half of Maxwell's equations remains

$$\partial_\mu \tilde{F}^{\mu\nu} = 0. \tag{2.57}$$

So the return of sources does not spoil covariance. On the contrary, it completes it. Charges and currents enter as a four-vector, and the field equations remain tensor equations. The way fields act on matter is Lorentz invariant, and the way matter generates fields is Lorentz invariant as well.

2.8 Summary

The lecture has a very definite arc. We begin with the Lorentz force law and identify the missing half of the subject: not how fields act on matter, but how matter determines fields. Maxwell's equations answer that question. In vacuum they can be written as the covariant equations $\partial_\mu F^{\mu\nu} = 0$ and $\partial_\mu \tilde{F}^{\mu\nu} = 0$, and from them we derive wave equations for the field components themselves. A plane-wave solution then shows that electromagnetic radiation is transverse and polarized, with $\mathbf{E}$, $\mathbf{B}$, and the direction of propagation mutually perpendicular.

Finally, when charges and currents are restored, the continuity equation and the four-current J^μ lead to the sourced covariant equation $\partial_\mu F^{\mu\nu} = J^\nu$. That is the closing point Susskind wants to leave with us: electromagnetism can be written entirely in Lorentz-covariant form, and because those same laws imply waves moving with speed c, light travels with the same speed in every inertial frame.